

Slutsky Decomposition: The Simple Version

Forget the Jargon. Here's What's Actually Happening:

You buy coffee and donuts every week. Suddenly, coffee gets more expensive.

Obviously, you're going to buy less coffee. But **WHY** are you buying less? There are actually **TWO separate reasons**:

Reason 1: "Coffee is now expensive compared to donuts"

You think: "Wow, coffee costs way more than donuts now. I should buy more donuts and less coffee because I get more bang for my buck with donuts."

→ This is the **SUBSTITUTION EFFECT**

Reason 2: "I'm poorer now"

You think: "Damn, my paycheck is the same but everything feels more expensive. I need to cut back on everything."

→ This is the **INCOME EFFECT**

That's it. That's the whole thing.

Concrete Example: You and Your Coffee

Initial Situation

- You have \$20/week to spend
- Coffee costs \$2 per cup
- Donuts cost \$2 each
- You buy: **5 coffees + 5 donuts** (spending all \$20)
- Your happiness level: Let's call it **100 happy points**

What Happens: Coffee Price Doubles to \$4

You end up buying: **2 coffees + 6 donuts**

So you went from 5 coffees down to 2 coffees. That's a change of **-3 coffees**.

But let's break down **WHY** you cut back by 3 coffees...

Breaking It Down

STEP 1: The Substitution Effect

Imagine I said: "Hey, I know coffee just got expensive. Let me give you extra money so you can still afford to be exactly as happy as before (100 happy points)."

With this extra compensation, you'd have enough money to stay at 100 happy points. But even with this extra cash, coffee is still *relatively expensive* compared to donuts now.

So you'd think: "Coffee is twice as expensive as donuts now (\$4 vs \$2). Even though I can afford to be just as happy, I should substitute toward donuts and away from coffee."

You'd buy: **4 coffees + 7 donuts** (with the extra money I gave you)

$$\text{Substitution Effect} = 4 - 5 = -1 \text{ coffee}$$

You cut back by 1 coffee just because of the *relative price change*, even though you had enough money to stay equally happy.

STEP 2: The Income Effect

Now I take away the extra money I gave you. You're back to your real \$20/week budget.

But remember, coffee is now \$4 instead of \$2. Your \$20 doesn't go as far anymore. You're effectively **poorer**.

Being poorer means you need to cut back on *everything* (both coffee and donuts).

You go from: **4 coffees** (with compensation) down to **2 coffees** (without compensation)

$$\text{Income Effect} = 2 - 4 = -2 \text{ coffees}$$

You cut back by another 2 coffees just because you're effectively poorer now.

STEP 3: Add Them Up

The Slutsky Decomposition

Total change in coffee = -3 coffees

This breaks down as:

Substitution Effect: -1 coffee (coffee is expensive relative to donuts)

Income Effect: -2 coffees (you're poorer overall)

Check: $-1 + (-2) = -3$

The Visual Version

Think of it like this:

Original: 5 coffees, 5 donuts, 100 happy points

↓ *Coffee price doubles, but I give you extra money to stay at 100 happy points*

Compensated: 4 coffees, 7 donuts, 100 happy points

This move is the **SUBSTITUTION EFFECT**: -1 coffee

↓ *I take away the extra money, back to real \$20 budget*

Final: 2 coffees, 6 donuts, maybe 60 happy points

This move is the **INCOME EFFECT**: -2 coffees

Key Insights

1. Substitution effect is **ALWAYS** negative (or zero)

- When price goes up, you always substitute away from that good
- It's about *relative prices*
- "Coffee vs donuts: which gives more happiness per dollar?"

2. Income effect depends on the type of good

- **Normal good** (like coffee): When you're poorer, you buy less
 - Income effect is negative
 - Both effects work together → demand definitely falls
- **Inferior good** (like instant ramen): When you're poorer, you buy MORE
 - Income effect is positive
 - The two effects fight each other

3. Special cases (you'll see these on the exam):

- **Perfect complements** (like left shoes and right shoes): Substitution effect = 0
 - You can't substitute! You always need them in fixed proportions
 - Entire change is income effect
- **Quasilinear utility**: Income effect = 0 for the "special" good
 - Entire change is substitution effect

How to Solve These Problems on the Exam

The Recipe (Simple Version)

Given: Price of good 1 rises from p_1 to p'_1

1. Find the original bundle

- Use the standard optimal choice formulas
- For Cobb-Douglas $u = x_1^a x_2^b$: spend $\frac{a}{a+b}$ of income on good 1
- Calculate the utility level at this bundle

2. Find the compensated bundle

- Stay at the same utility level
- But use the NEW price ratio
- Mathematically: solve $MRS = \frac{p'_1}{p_2}$ while keeping $u = \bar{u}$

3. Find the new bundle

- Use the NEW prices
- Use the ORIGINAL income
- Just solve the regular problem again

4. Calculate the effects

- Sub effect = compensated - original
- Income effect = new - compensated
- Total = new - original (should equal sub + income)

Cobb-Douglas Shortcut

For $u = x_1 x_2$ with equal exponents:

- **Original:** Always spend half your income on each good

$$- x_1^* = \frac{m}{2p_1}, x_2^* = \frac{m}{2p_2}$$

- **Compensated:** The $MRS = p'_1/p_2$ gives you $\frac{x_2}{x_1} = \frac{p'_1}{p_2}$

- Combine this with $x_1 x_2 = \bar{u}$ (the original utility)
- Solve for x_1 and x_2

- **New:** Spend half your income on each good again

$$- x_1^{new} = \frac{m}{2p'_1}, x_2^{new} = \frac{m}{2p_2}$$

Bottom Line

When a price changes, your consumption changes for two reasons:

1. You substitute toward the relatively cheaper good (substitution effect)
2. You're effectively richer or poorer, so you adjust everything (income effect)

The Slutsky decomposition just separates these two effects so you can see how much of the change came from each reason.

Remember:

- Three bundles: Original, Compensated, New
- Compensated = "What if I kept you equally happy at new prices?"
- Sub effect = Always opposes price change
- Income effect = Depends on normal vs inferior
- Total = Sub + Income (always!)